

Machine Learning I — Hands-On 10

10.1 From transactions to itemsets

In association rule mining, the data are often represented as a set of **transactions**. Each transaction contains a set of items. For example, in market basket analysis, each transaction corresponds to one customer purchase, and the items correspond to the products bought together.

Consider the following set of supermarket transactions.

Transaction ID	Items bought
T1	Bread, Milk, Butter
T2	Bread, Milk
T3	Bread, Eggs
T4	Milk, Butter
T5	Bread, Milk, Butter, Eggs
T6	Bread, Butter
T7	Milk, Eggs
T8	Bread, Milk, Butter
T9	Bread, Milk, Coffee
T10	Coffee, Eggs

Use this transaction dataset to compute itemset supports.

The **support** of an itemset X is the proportion of transactions that contain all items in X :

$$\text{support}(X) = \frac{\#\{t : X \subseteq t\}}{N}.$$

1. Compute the support of the following itemsets:

- {Bread};
- {Milk};
- {Butter};
- {Eggs};
- {Coffee}.

2. Compute the support of the following itemsets:

- {Bread, Milk};
- {Bread, Butter};
- {Milk, Butter};
- {Bread, Milk, Butter};
- {Coffee, Eggs}.

3. Assume a minimum support threshold of

$$\text{minsup} = 30\%.$$

Identify which itemsets from Questions 1 and 2 are **frequent**.

4. Explain why support measures the frequency of an itemset, but does not by itself describe the strength of an implication such as

$$\{\text{Bread}\} \rightarrow \{\text{Milk}\}.$$

10.2 From itemsets to association rules

Frequent itemsets can be used to construct **association rules**. An association rule has the form

$$X \rightarrow Y, \quad X \cap Y = \emptyset,$$

where X is the antecedent and Y is the consequent.

The support of a rule is the support of the union of the antecedent and consequent:

$$\text{support}(X \rightarrow Y) = \text{support}(X \cup Y).$$

The **confidence** of a rule is

$$\text{confidence}(X \rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}.$$

Confidence measures the proportion of transactions containing X that also contain Y .

Using the transaction dataset from Exercise 10.1, compute the support and confidence of the following rules.

1. $\{\text{Bread}\} \rightarrow \{\text{Milk}\}$
2. $\{\text{Milk}\} \rightarrow \{\text{Bread}\}$
3. $\{\text{Bread}\} \rightarrow \{\text{Butter}\}$
4. $\{\text{Butter}\} \rightarrow \{\text{Bread}\}$
5. $\{\text{Bread, Milk}\} \rightarrow \{\text{Butter}\}$
6. $\{\text{Coffee}\} \rightarrow \{\text{Eggs}\}$
7. Compare the following two rules:

$$\{\text{Bread}\} \rightarrow \{\text{Milk}\}, \quad \{\text{Milk}\} \rightarrow \{\text{Bread}\}.$$

8. Do they have the same support? Do they have the same confidence? Explain why.

10.3 Selecting rules using support and confidence

An association-rule mining task usually defines a minimum support and a minimum confidence. The goal is to find rules satisfying **both** thresholds:

$$\text{support}(X \rightarrow Y) \geq \text{minsup}, \quad \text{confidence}(X \rightarrow Y) \geq \text{minconf}.$$

This corresponds to the usual problem definition: given a transaction dataset, a minimum support, and a minimum confidence, obtain all rules that satisfy both conditions.

Consider the thresholds

$$\text{minsup} = 30\%, \quad \text{minconf} = 70\%.$$

Use the rules from Exercise 10.2.

1. For each rule, decide whether it satisfies the minimum support threshold.
2. For each rule, decide whether it satisfies the minimum confidence threshold.
3. Identify which rules are valid association rules under **both** thresholds.
4. Suppose we increase the minimum confidence threshold to

$$\text{minconf} = 90\%.$$

Which rules remain valid?

5. Explain what happens when the minimum confidence threshold becomes stricter.

10.4 Why confidence can be misleading

A high-confidence rule is not always interesting. A rule may have high confidence simply because the consequent is already very common.

To evaluate whether a rule is more informative than expected, we can use **lift**:

$$\text{lift}(X \rightarrow Y) = \frac{\text{confidence}(X \rightarrow Y)}{\text{support}(Y)}.$$

- If $\text{lift} > 1$, X and Y occur together more often than expected under independence.
- If $\text{lift} = 1$, X and Y are approximately independent.
- If $\text{lift} < 1$, X and Y occur together less often than expected under independence.

Using the transaction dataset from Exercise 10.1:

1. Compute the lift of $\{\text{Bread}\} \rightarrow \{\text{Milk}\}$.
2. Compute the lift of $\{\text{Milk}\} \rightarrow \{\text{Bread}\}$.
3. Compute the lift of $\{\text{Coffee}\} \rightarrow \{\text{Eggs}\}$.
4. Compute the lift of $\{\text{Eggs}\} \rightarrow \{\text{Coffee}\}$.
5. For each rule, state whether the association is positive, approximately independent, or negative.
6. Based on support, confidence, and lift, which of the above rules would you consider the most interesting? Justify your answer. Your answer should make explicit **which notion of interestingness** you are prioritizing: frequency, reliability of implication, or departure from independence.